

Domoic acid and human exposure risks: a review.

Domoic acid is a potent neurotoxin that is naturally produced by several diatom species of the genus *Pseudo-nitzschia*. The toxin acts as a glutamate agonist and is excitotoxic in the vertebrate central nervous system and other glutamate receptor-rich organs. Human exposure to domoic acid occurs via the consumption of contaminated shellfish that have accumulated the toxin while filter feeding on toxigenic phytoplankton during blooms. The first reported human domoic acid poisoning event occurred in Canada in 1987 during which clinical signs of acute toxicity such as gastrointestinal distress, confusion, disorientation, memory loss, coma and death were observed. The illness was named amnesic shellfish poisoning (ASP) and due to effective seafood monitoring programs there have been no documented ASP cases since 1987. However, domoic acid poisoning has a significant effect on marine wildlife and multiple poisoning events have occurred in marine birds and mammals over the last few decades. Currently, domoic acid producing diatom blooms are thought to be increasing in frequency world wide, posing an increasing threat to wildlife and human health. Of particular concern are the potential impacts of long-term low-level exposure in "at risk" human populations. The impacts of repetitive low-level domoic acid exposure are currently unknown. This review provides a basic description of the mechanism of action of domoic acid as well as a synthesis of information pertaining to domoic acid exposure routes, toxin susceptibility, and the importance of effective monitoring programs. The importance of investigating the potential human health impacts of long-term low-level domoic acid exposure in "at risk" human populations is also discussed. Published by Elsevier Ltd.